

X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-09-13

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:

- CSP Partial cross section data
- CST Temperature dependent cross section
- ETA Average neutron yield per nonelastic event
- NU Fission-neutron yield, nu-bar
- TKE Total kinetic energy
- SIG Cross section data given with limit as DATA and DATA-MIN
- SIG Production cross section, where product coded as SF4 or in DATA (ELEM/MASS)
- PFNS Prompt-fission neutron energy spectra

-- Currently implemented search: by full-Reaction-code, DatasetID, Author1, product

2. Plotting partial cross sections from local EXFOR and remote ENDF databases

- CSP vs ENDF MT:51/MF:3,33 - Production of a neutron, with residual in the 1st excited state
- NU vs ENDF MT452,455,456/MF:1,31 - Average number of total/delayed/prompt neutrons released per fission event
- ETA vs ENDF MT:452,18,102/MF:1,3,31,33 - Average neutron yield per nonelastic event: $\eta \pm \Delta\eta$
- PFNS vs ENDF - Prompt-fission neutron energy spectra (ratio to Maxwellian: $T=1.32\text{MeV}$)

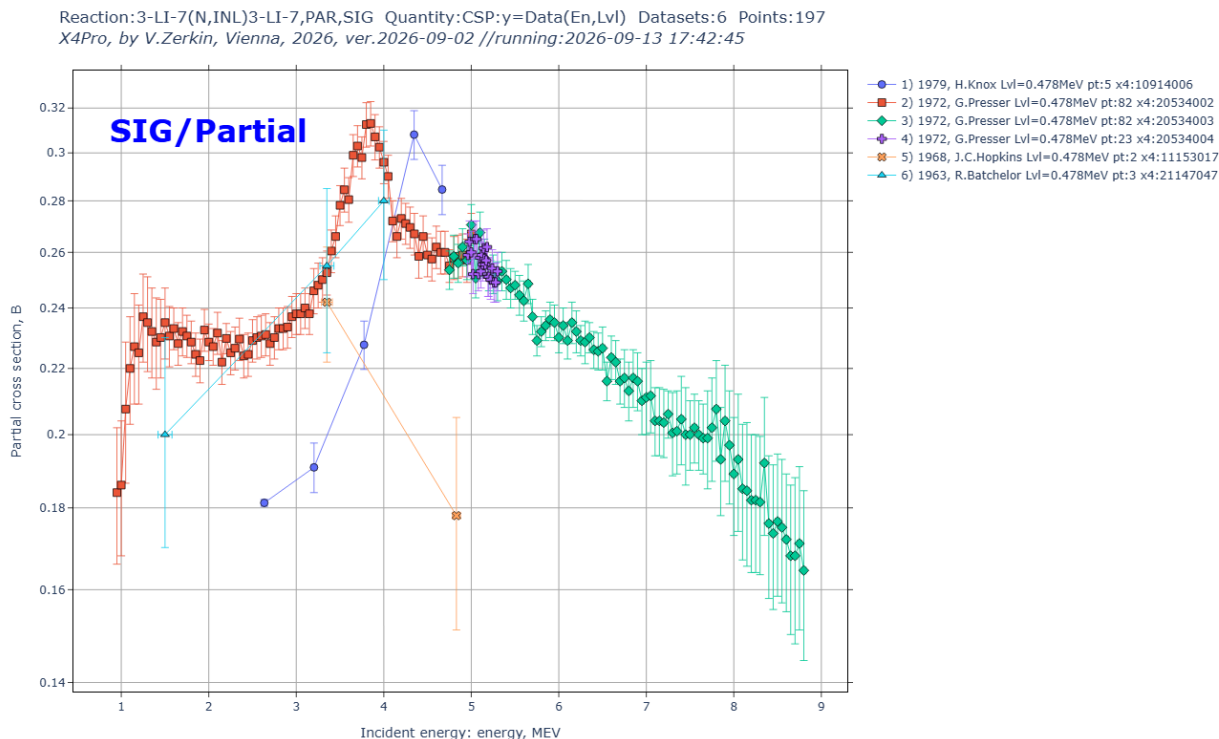
Code reac1.py help:

```

Program: reac1.py, ver.2026-08-14
Author: V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
* computational data are presented as function y=f(x1,x2,...)
  where f is measured quantity, xi are independent variables:
  EN - incident energy, E2 - energy of outgoing particle,
  ANG - angle of outgoing particle, EVL - energy level, etc.
Algorithm:
* user defines EXFOR Reaction-code for data search, x-variable by number
* selects x-variable by number (default: x1)
* other variables will be used as parameter for grouping of data points
Usage:
$ python [{flag}] x4update.py [{option|reaction}]
* flag: see all Python flags: $ python --help
      -B don't write .pyc files on import
* option:
  -help print this help-text and exit (also --help)
  -o:<file> output file name
  -x:x? set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
  -x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
  -x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
  -x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
  -nogrp avoid grouping datasets by reaction-codes on the plot
  -fx:<num> set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
  -fy:<num> set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
plotting:
  -xlog logarithmic scale of X-Axis
  -ylog logarithmic scale of Y-Axis
  -lines connect points on the plot by lines
  -sym draw border of a symbol of a data point on the plot
  -annot:x,y,txt annotation on top of the plot;
           x and y should be given in units of the plot;
           text can include html tags: <b>, <sup>
* reaction:
  <reaction> full EXFOR Reaction-code, e.g. "8-0-16(N,EL)8-0-16,,DA"
  
```

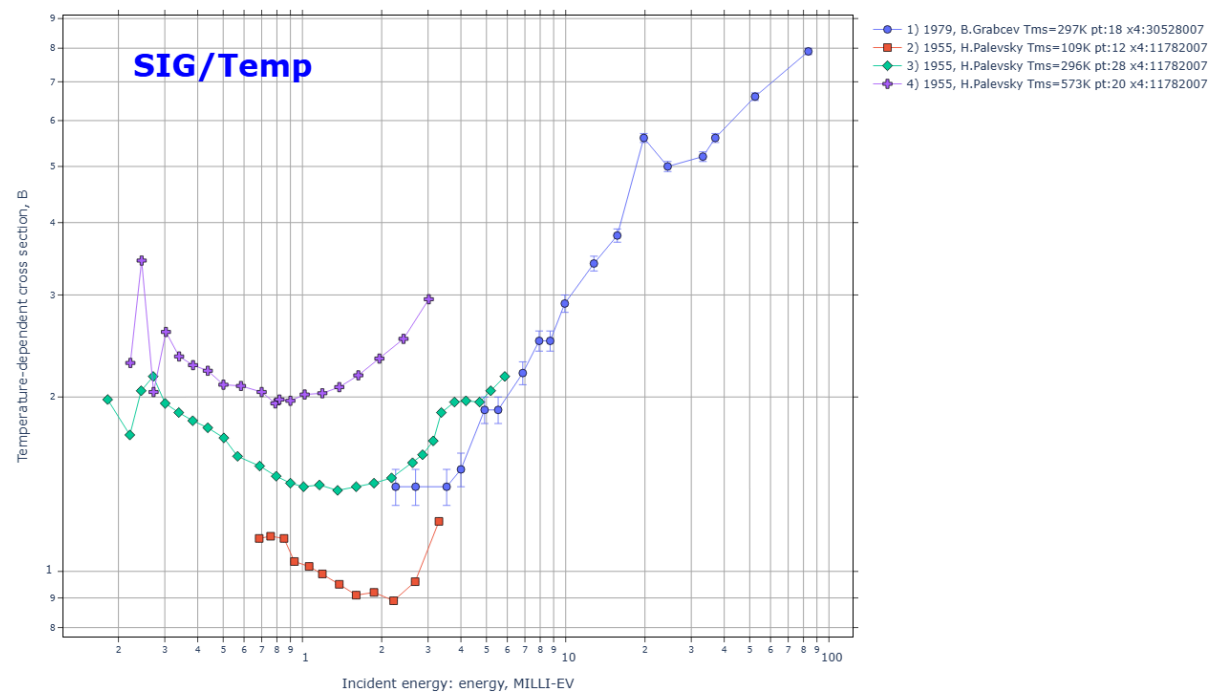
part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data



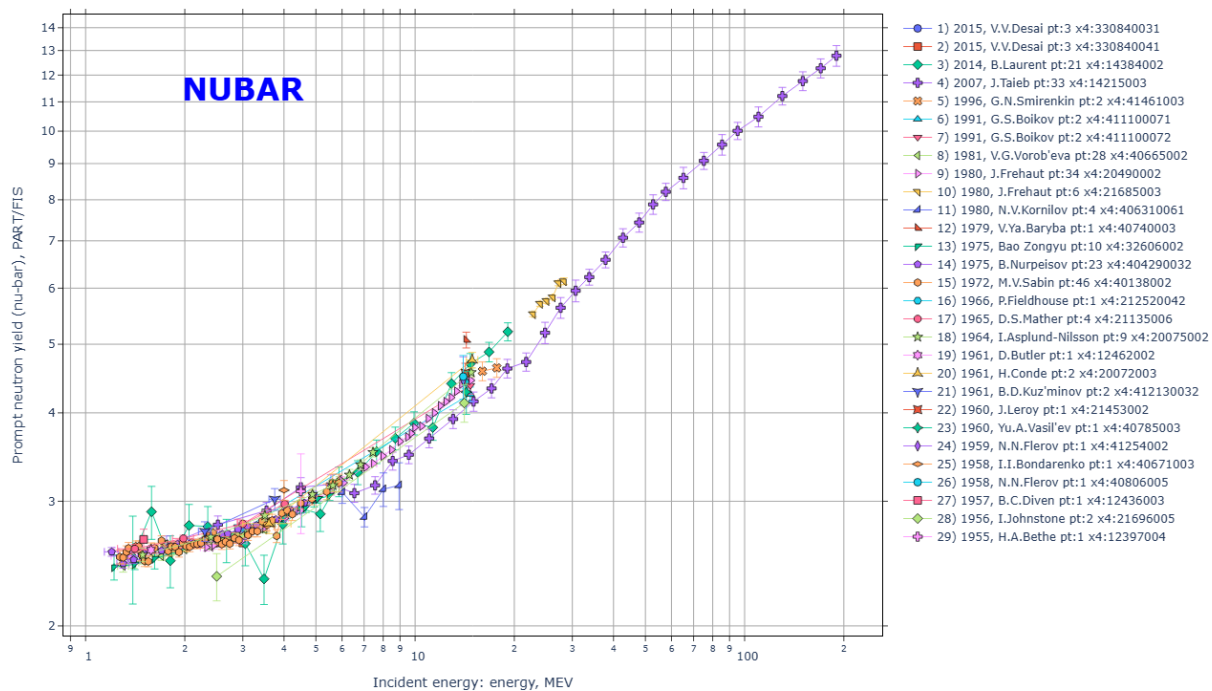
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section

Reaction: 82-PB-0(N,TOT),,SIG/TMP Quantity: CST:y=Data(En,Tms) Datasets: 4 Points: 78
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:33

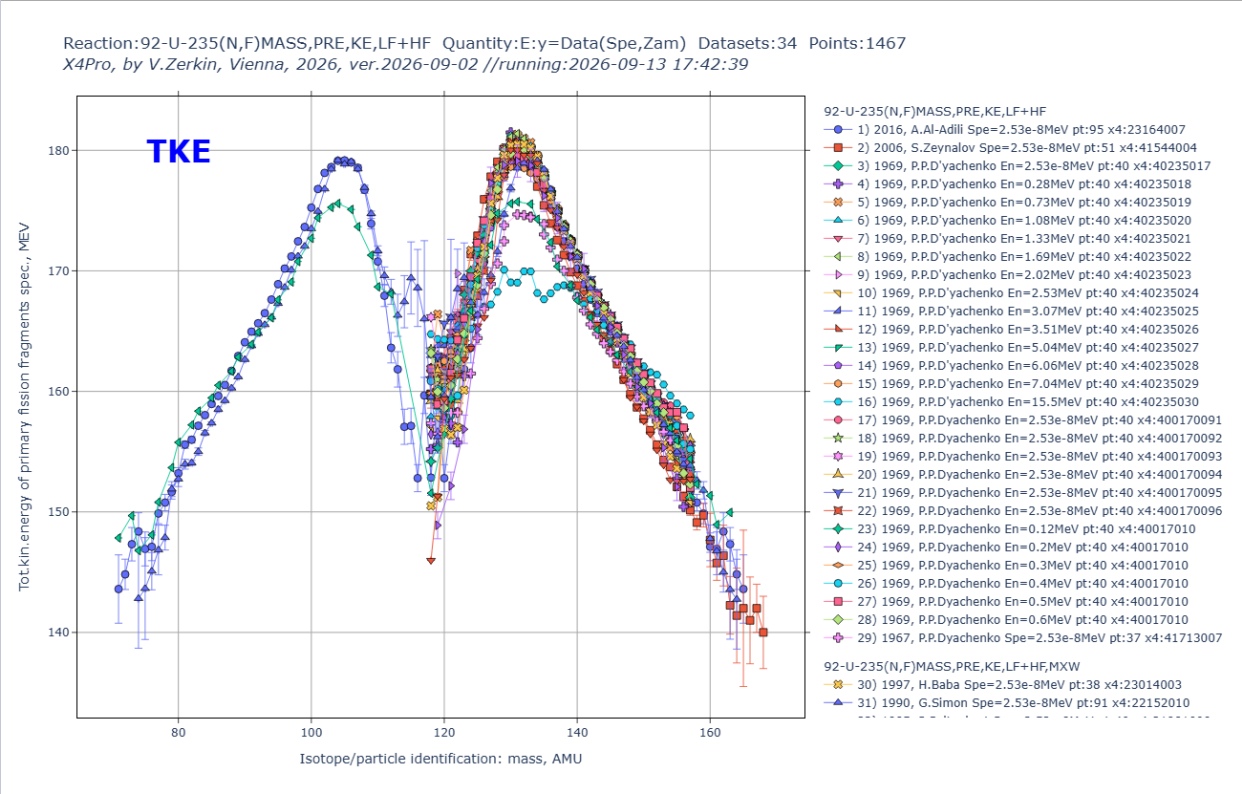


3) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar

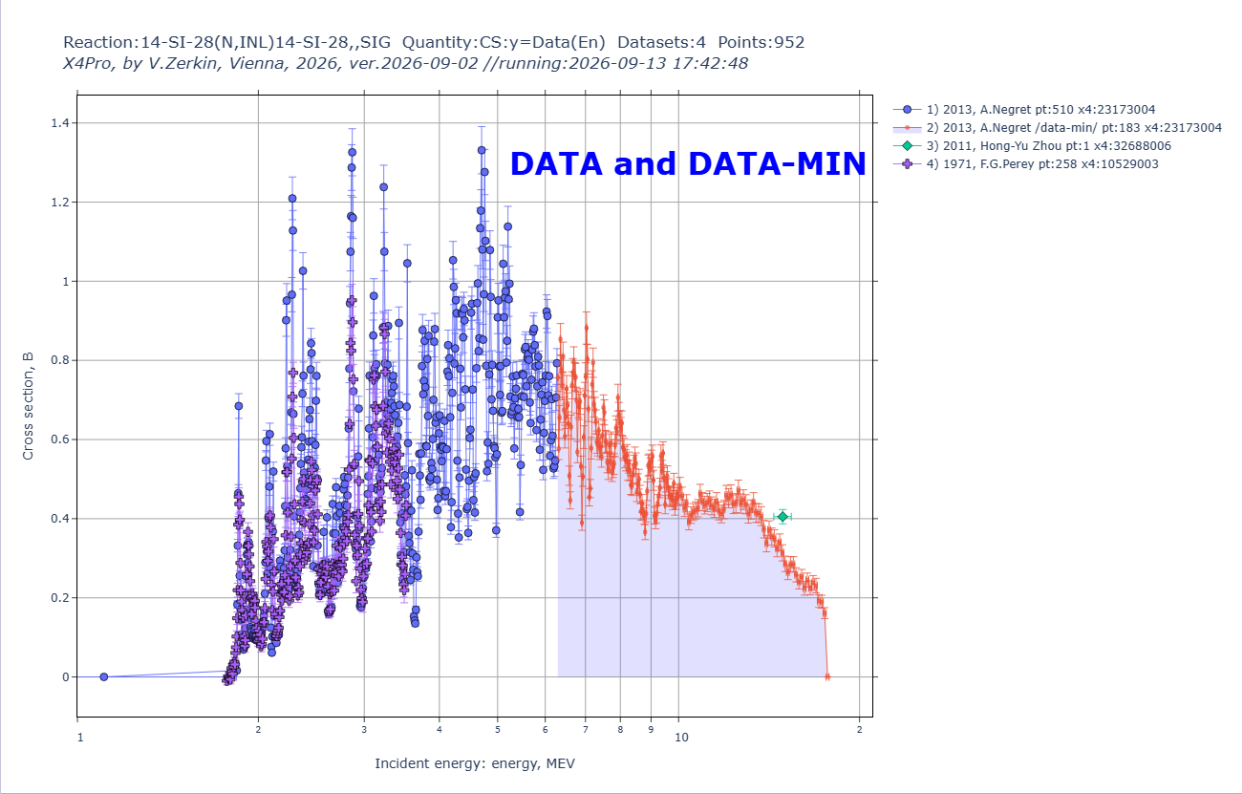
Reaction: 92-U-238(N,F),PR,NU Quantity: NU:y=Data(En) Datasets: 29 Points: 246
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:36



4) [part1-8-reac/out00/tke1.html.png](#) TKE Total kinetic energy

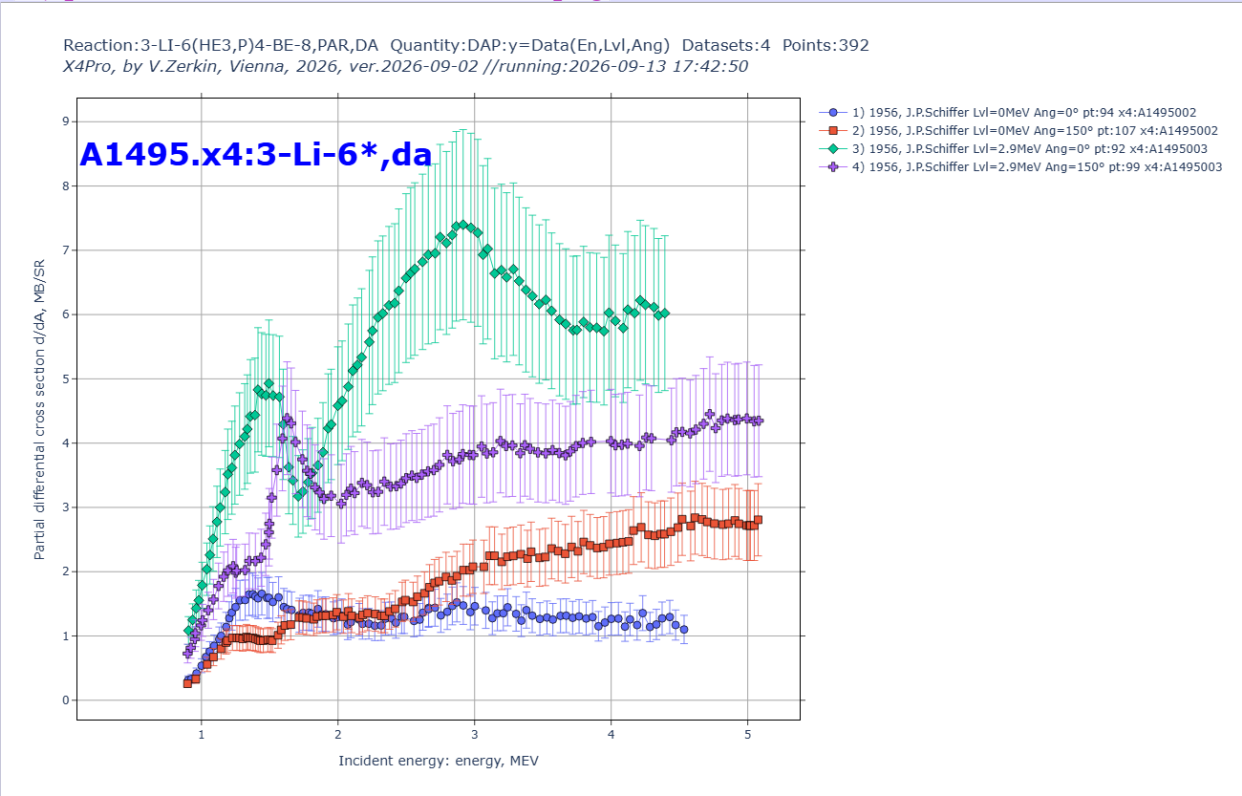


5) [part1-8-reac/out00/si28nn.html.png](#) SIG Cross section data given with limit as DATA and DATA-MIN



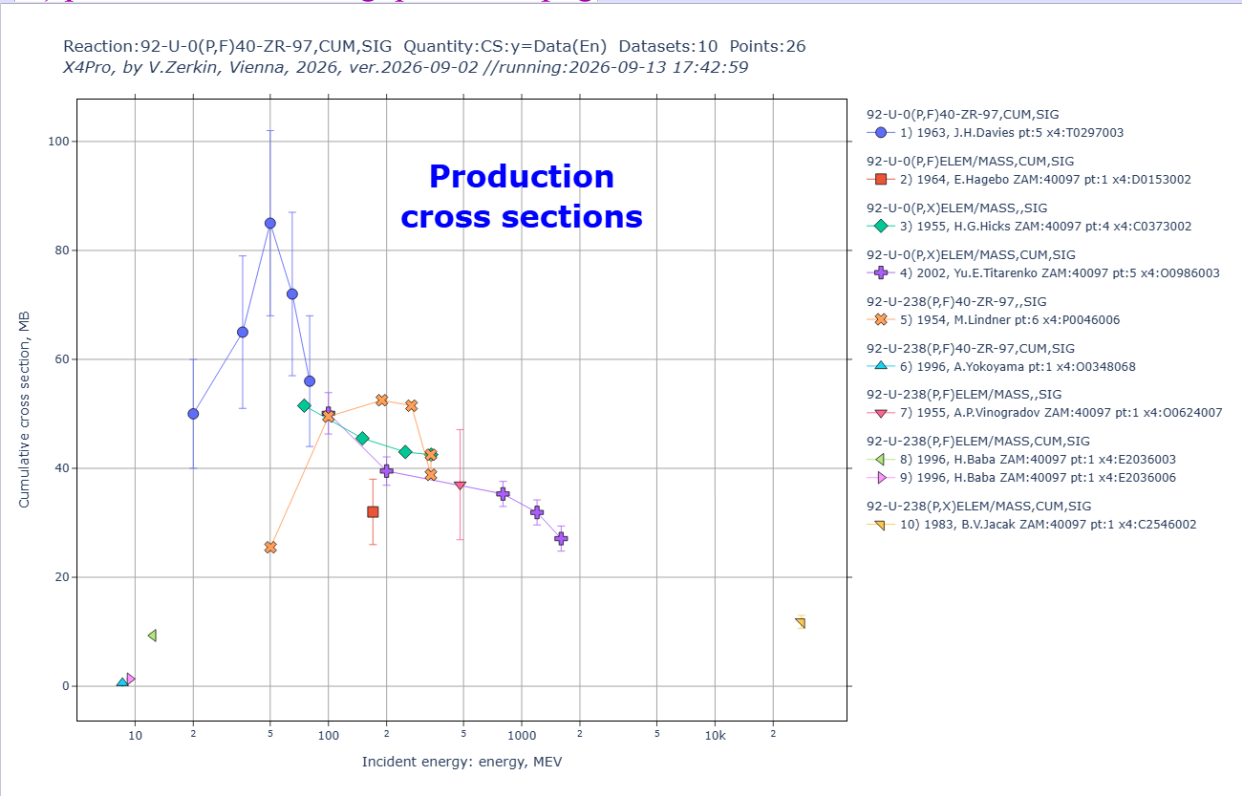
6) part1-8-reac/out00/a1495-li6.html.png

Data from EXFOR:"A1495" & Reaction: "3-Li-6*DA"



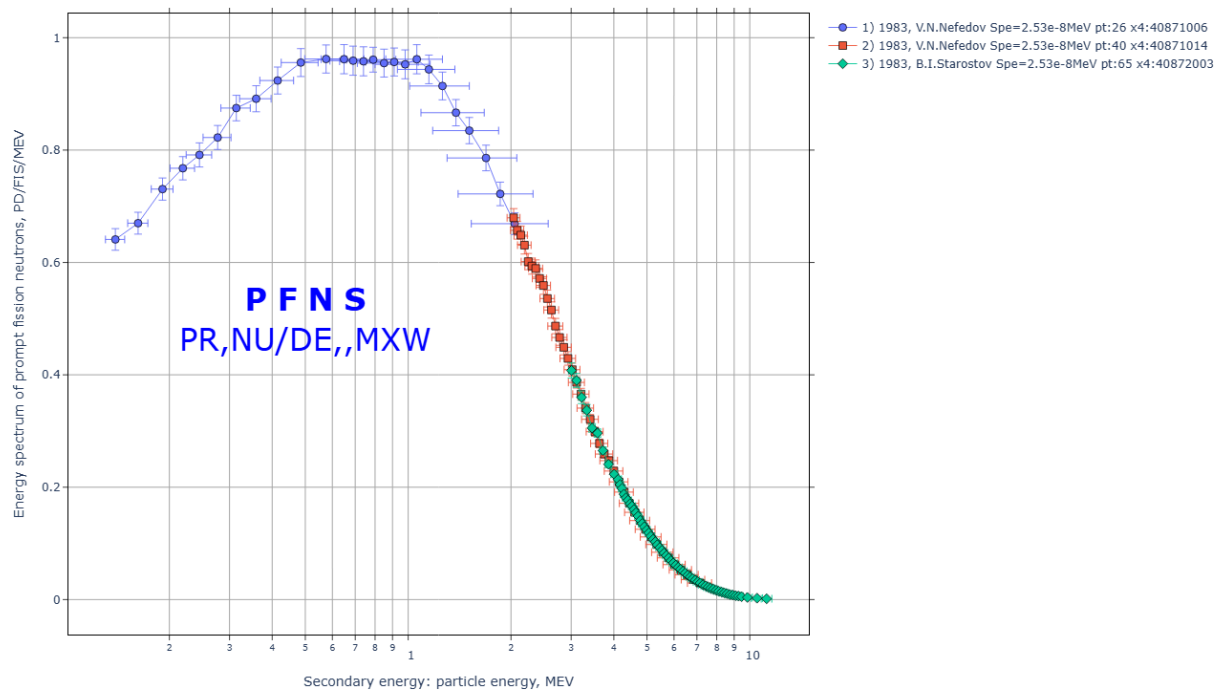
7) part1-8-reac/out00/sig1prod.html.png

Production cross section coded as SF4 and ELEM/MASS



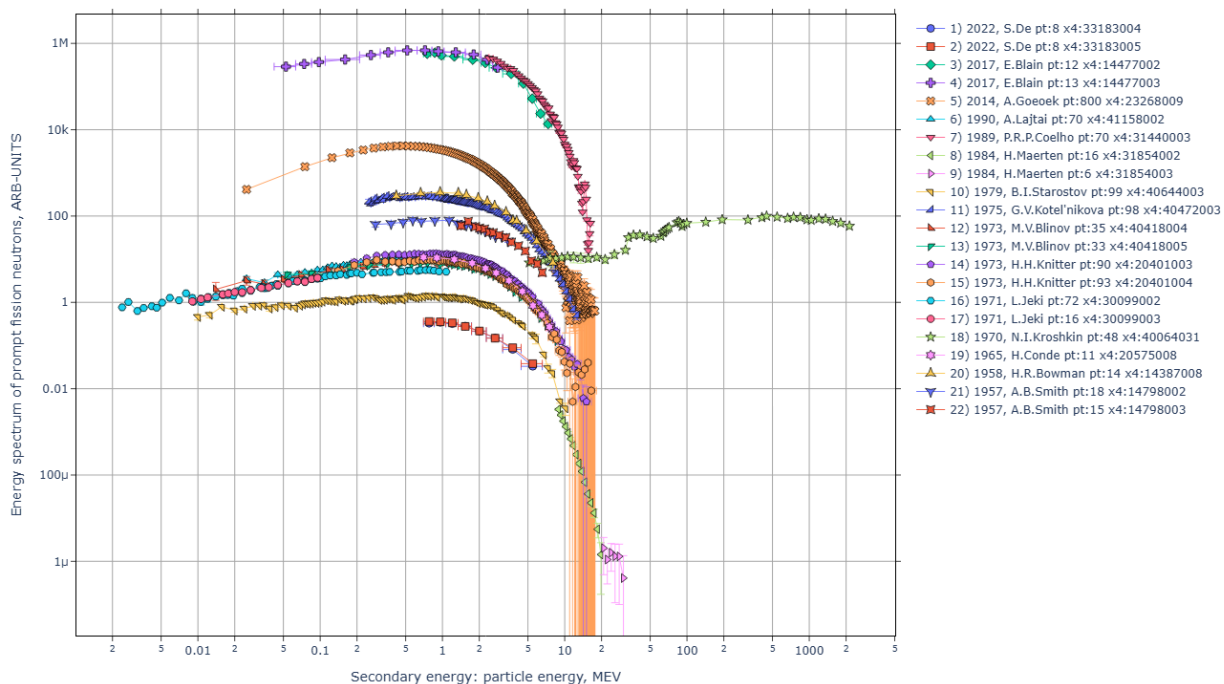
8) [part1-8-reac/out00/pfns1.html.png](#) PFNS Prompt-fission neutron energy spectra

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:43:01



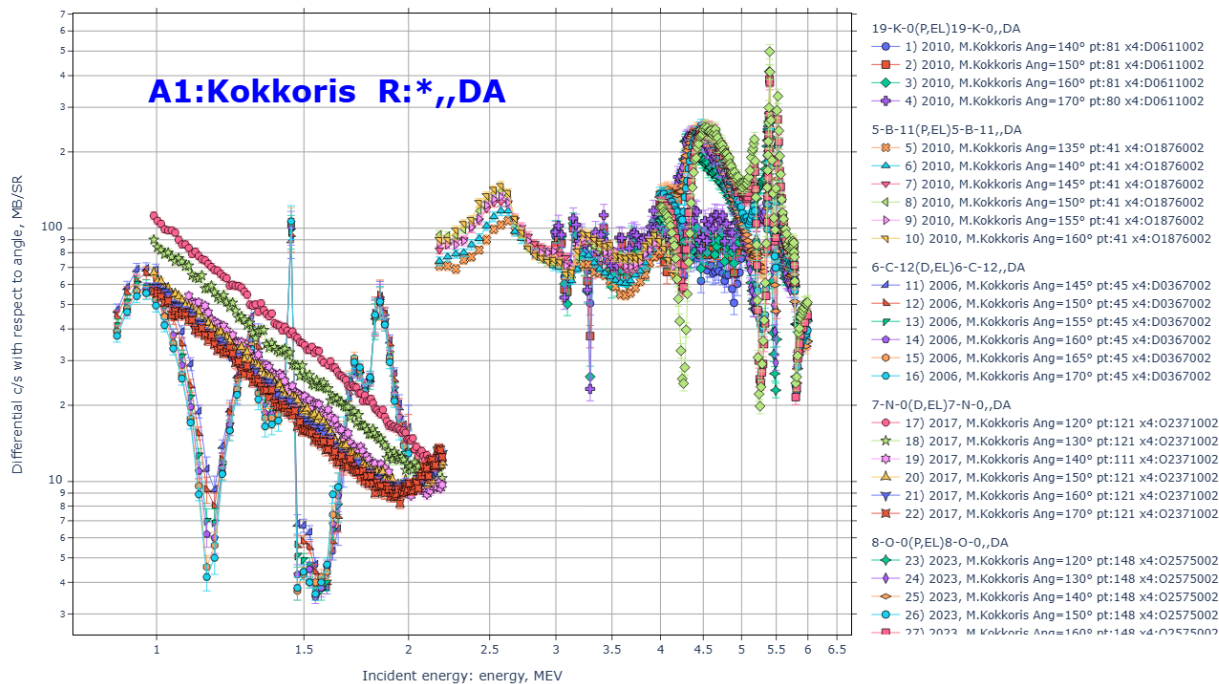
9) [part1-8-reac/out00/pfns4.html.png](#) Energy spectrum of prompt fission neutrons: relative data

Reaction: 98-CF-252(0,F),PR,NU/DE,,REL Quantity: MFQ:y=Data(E2) Datasets: 22 Points: 1645
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:43:09



10) part1-8-reac/out00/Kokkoris.html.png Data by "Kokkoris" & Reaction: "*,DA"

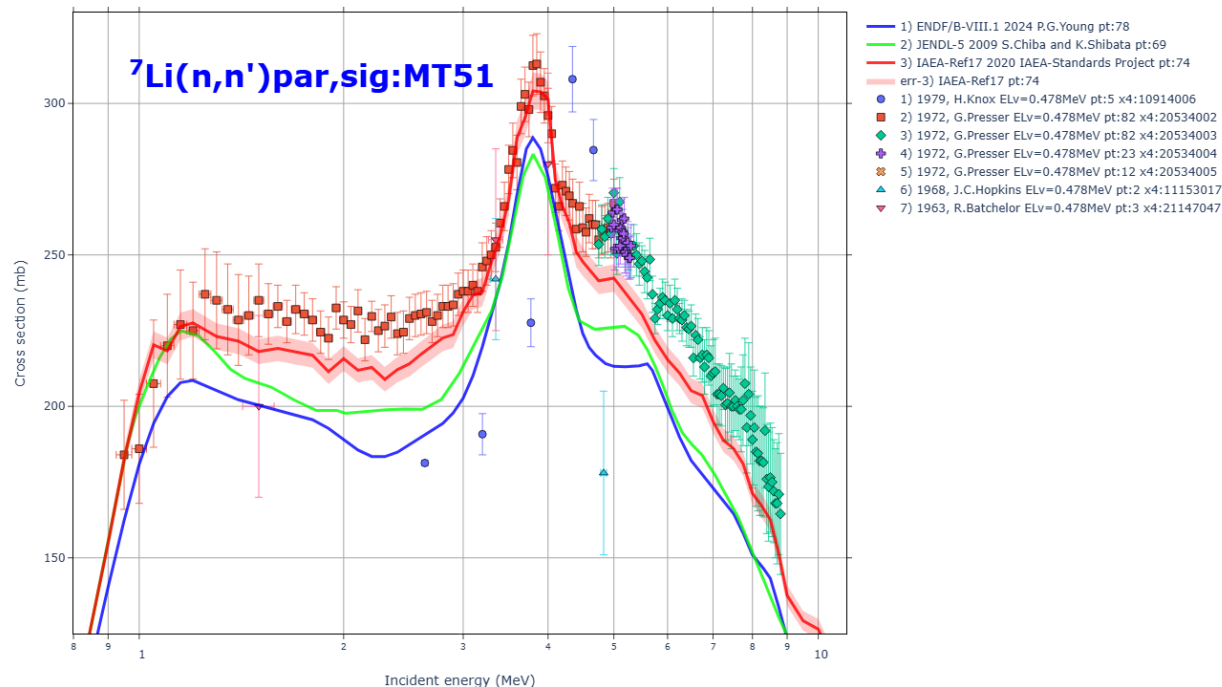
Reaction: 19-K-0(P,EL)19-K-0,,DA Quantity: DA:y=Data(En,Ang) Datasets: 28 Points: 2443
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:56



part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

11) part2-8-sig1par/out00/Li7ninl.html.png Production of a neutron, with residual in the 1st excited state

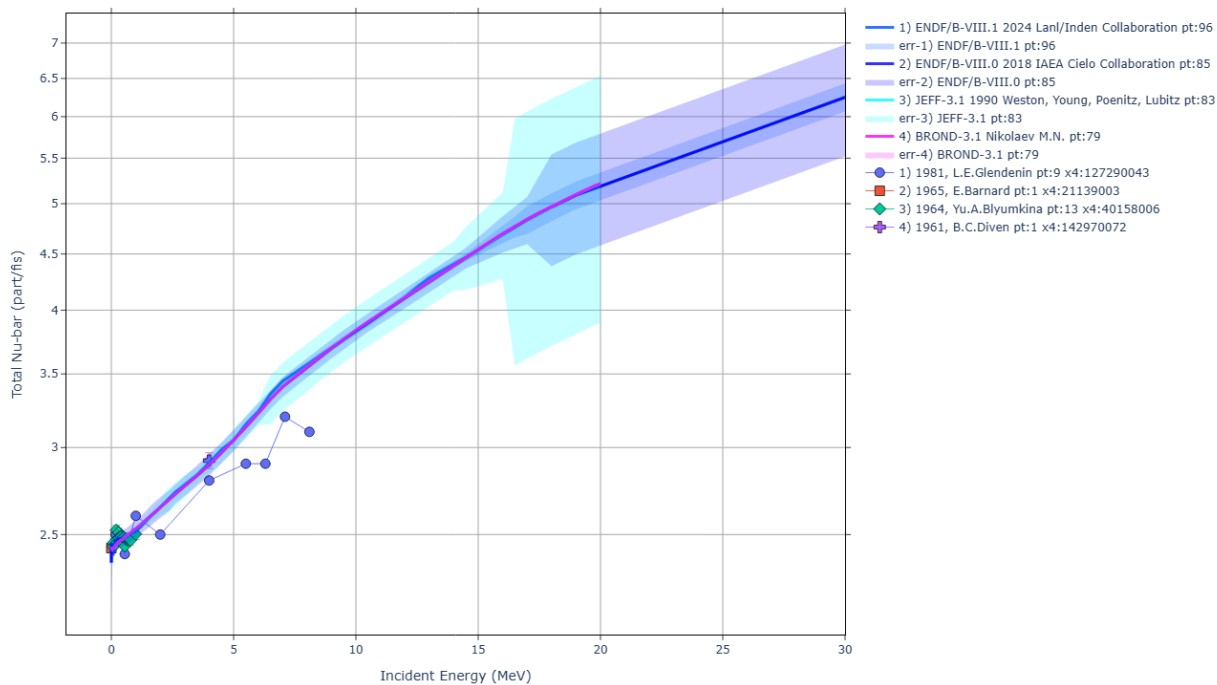
EXFOR/ENDF cross sections SIG(E): Li-7(n,inl)par,sig EXFOR-datasets: 7
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2026, ver.2026-08-10 //run:2026-09-13 17:47:04



part2-9-nubar/ EXFOR-ENDF NUBAR - average number of neutrons per fission

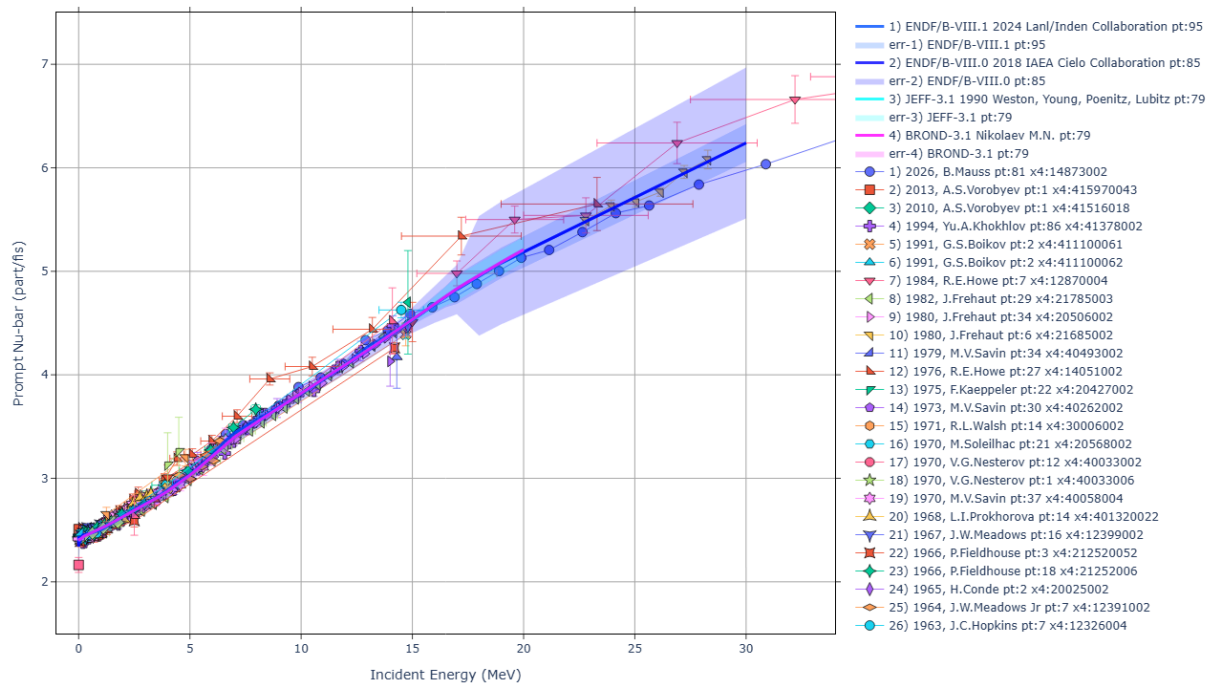
12) part2-9-nubar/out00/U235nf_-nu.html.png Total neutron yield (nu-bar)

EXFOR-ENDF NUBAR: 92-U-235(N,F),,NU Exp.Datasets:4 Points:24
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-13 17:47:51

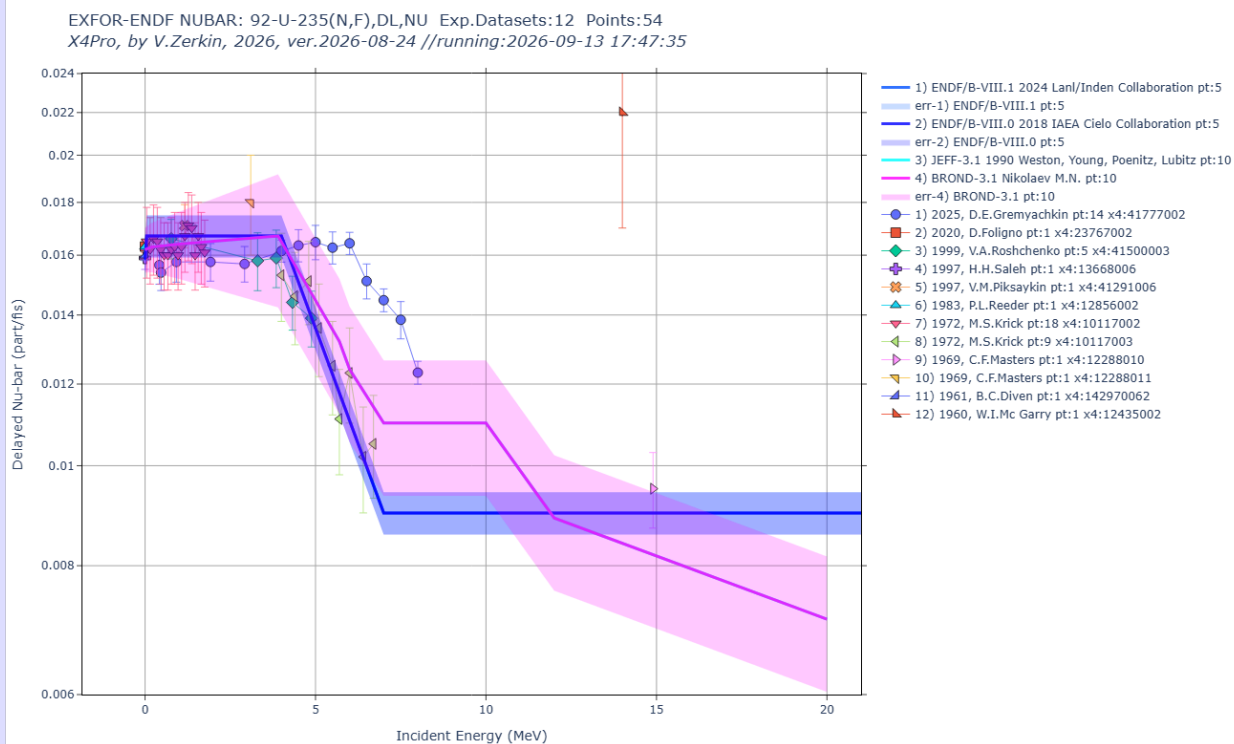


13) part2-9-nubar/out00/U235nf_pr-nu.html.png Prompt neutron yield (nu-bar)

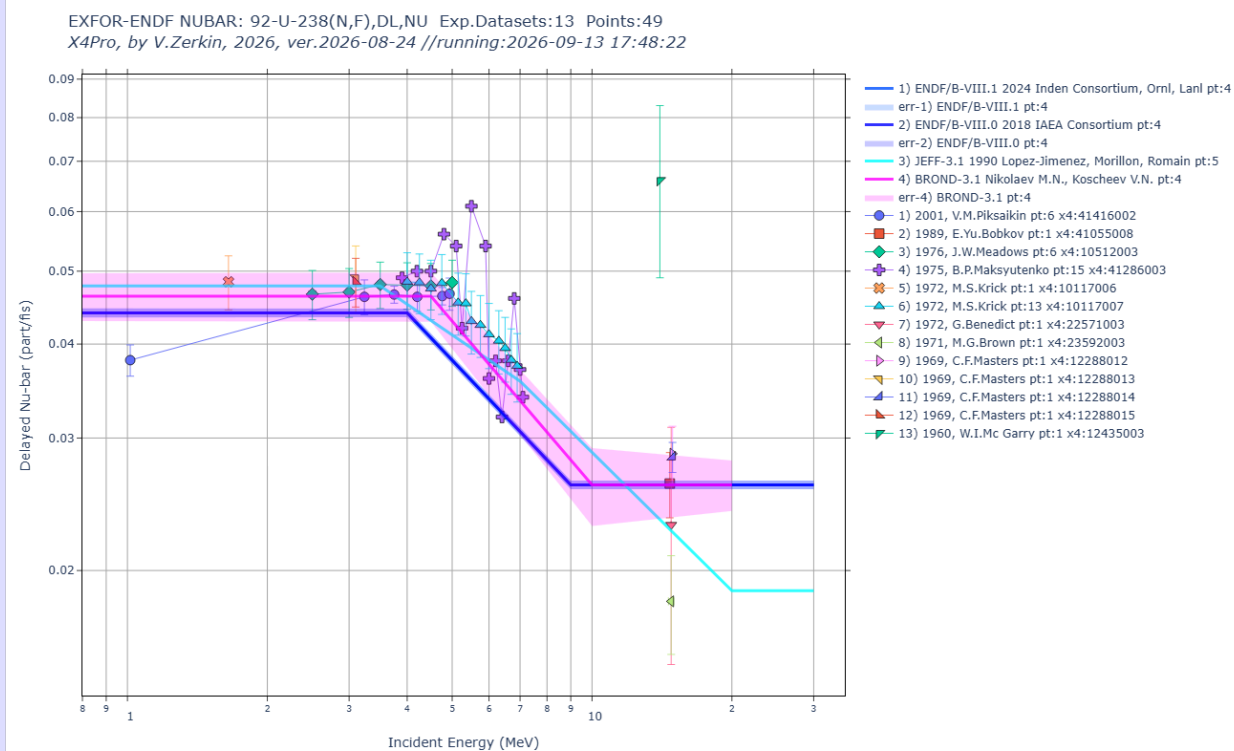
EXFOR-ENDF NUBAR: 92-U-235(N,F),PR,NU Exp.Datasets:38 Points:535
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-13 17:47:17



14) [part2-9-nubar/out00/U235nf_dl-nu.html.png](#) Delayed neutron yield



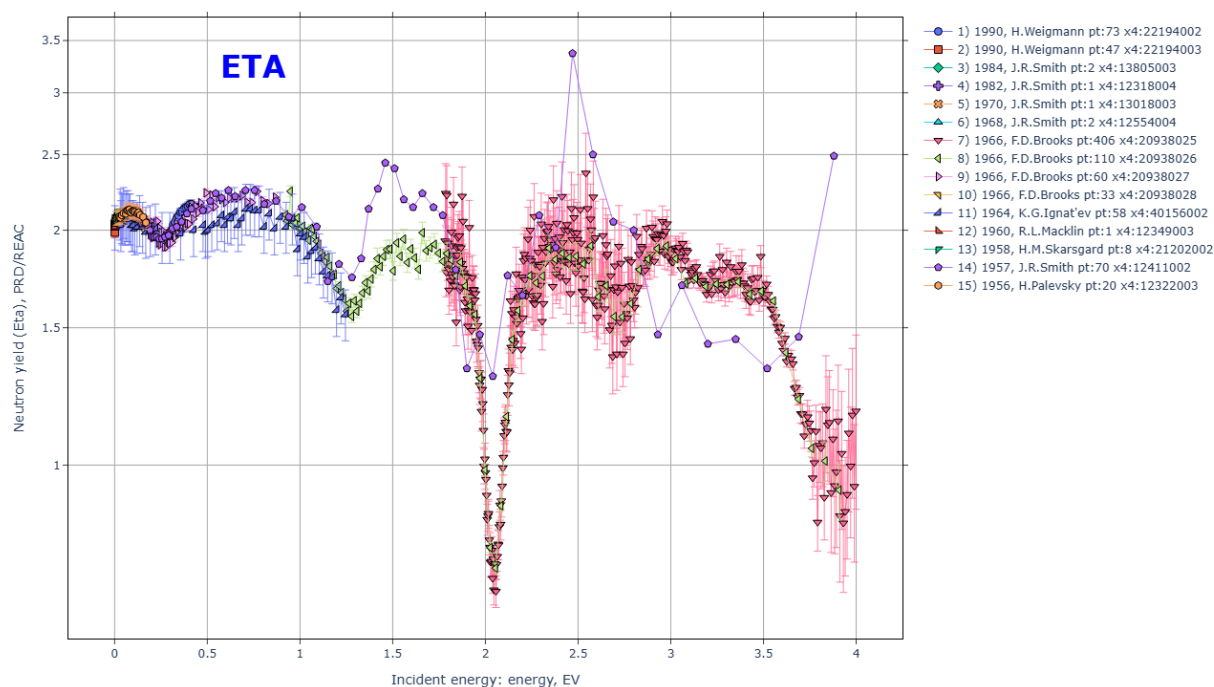
15) [part2-9-nubar/out00/U238nf_dl-nu.html.png](#)



part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

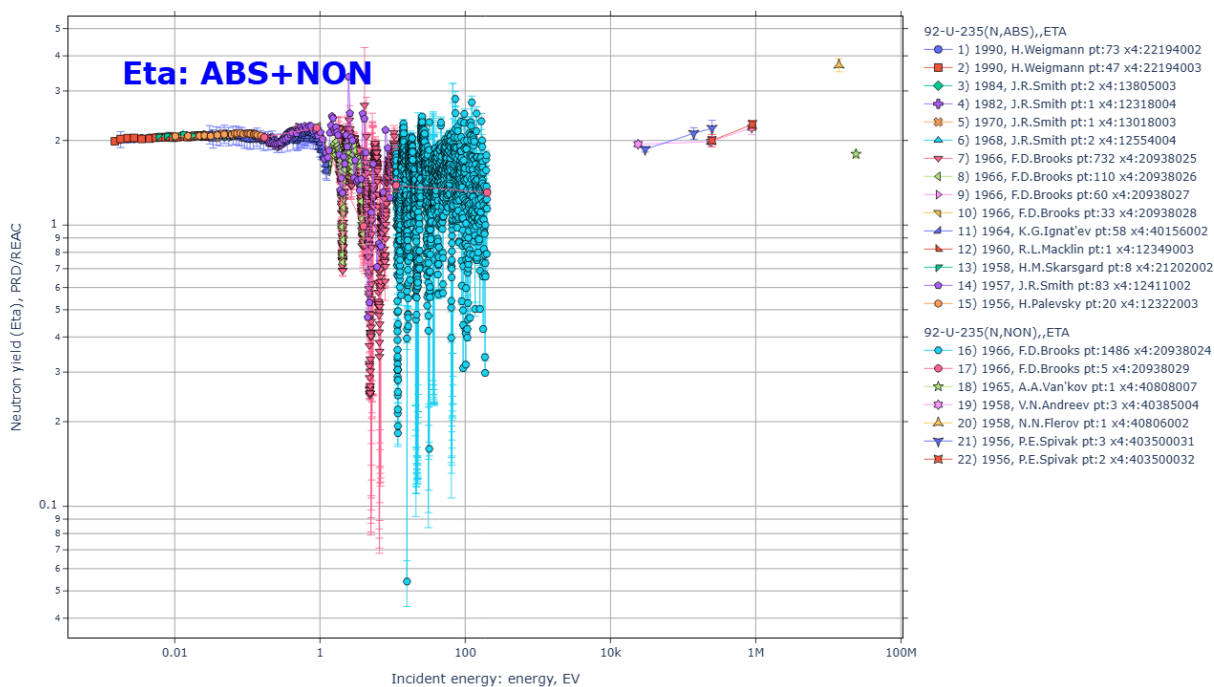
16) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

Reaction: 92-U-235(N,ABS),,ETA Quantity: NU:y=Data(En) Datasets: 15 Points: 892
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-08-28 13:26:55



17) part1-8-reac/out00/eta2.html.png ETA: (N,ABS)+(N,NON)

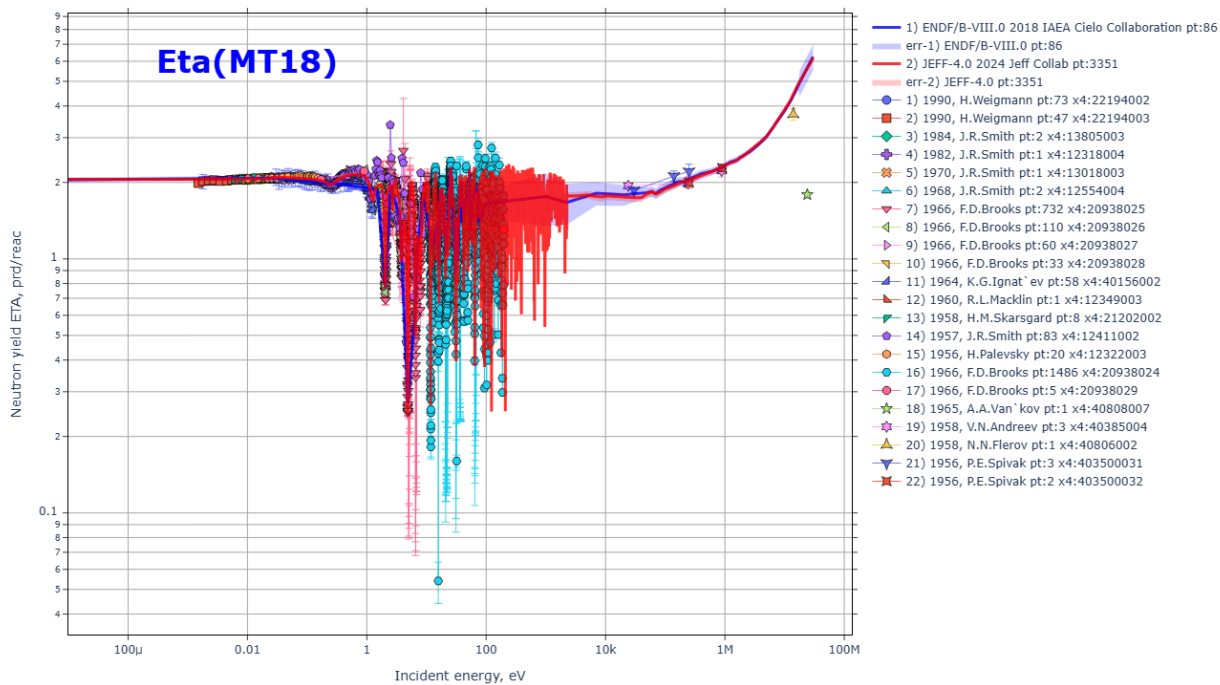
Reaction: 92-U-235(N,ABS),,ETA Quantity: NU:y=Data(En) Datasets: 22 Points: 2732
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:41



part2-A-eta/ EXFOR/ENDF Eta: neutron yield per nonelastic even

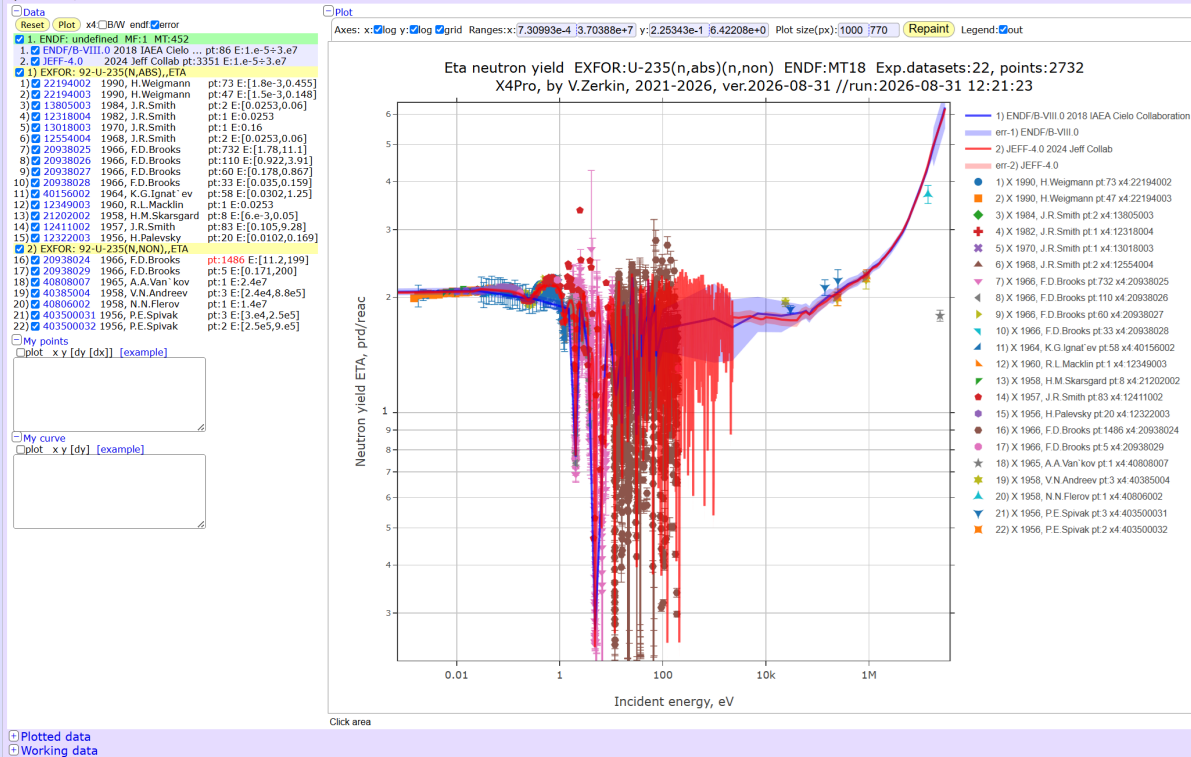
18) part2-A-eta/out00/U235-eta.html.png ETA: EXFOR vs. ENDF $\eta \pm \Delta\eta$

Eta neutron yield EXFOR:U-235(n,abs)(n,non) ENDF:MT18 Exp.datasets:22, points:2732
 X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-13 17:48:53



19) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT:452,18,102,MF:1,3,31,33

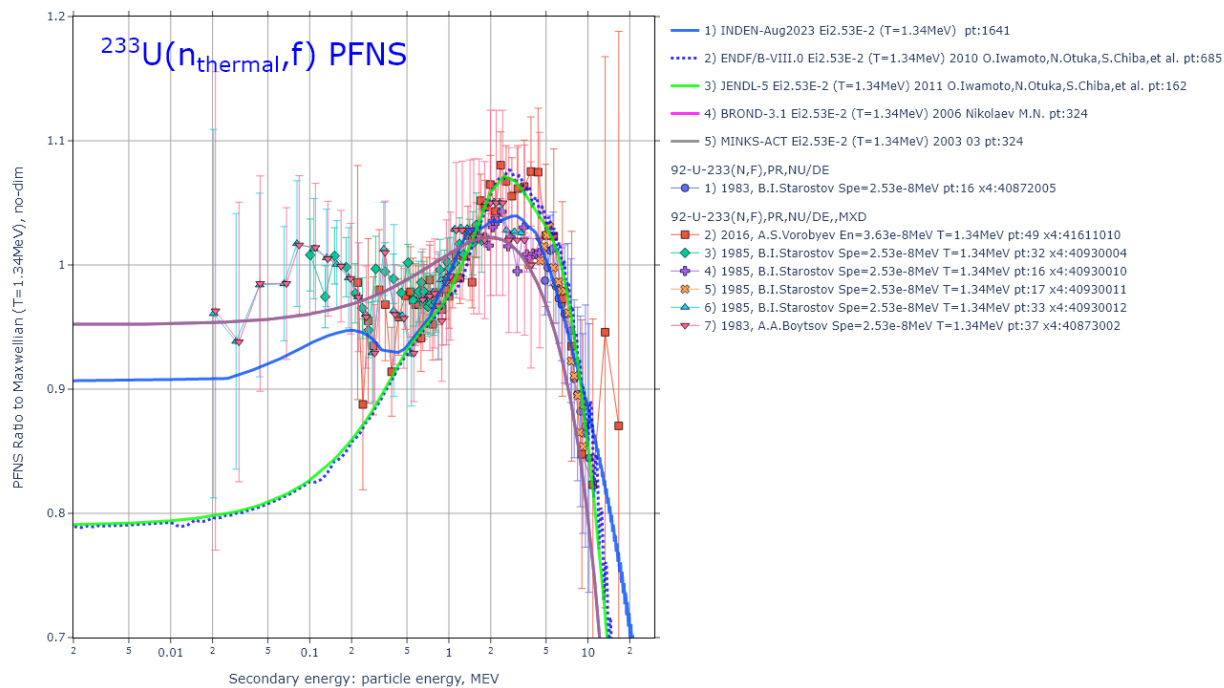
X4Pro - advanced plot
 by V.Zerkin, 2025-2026



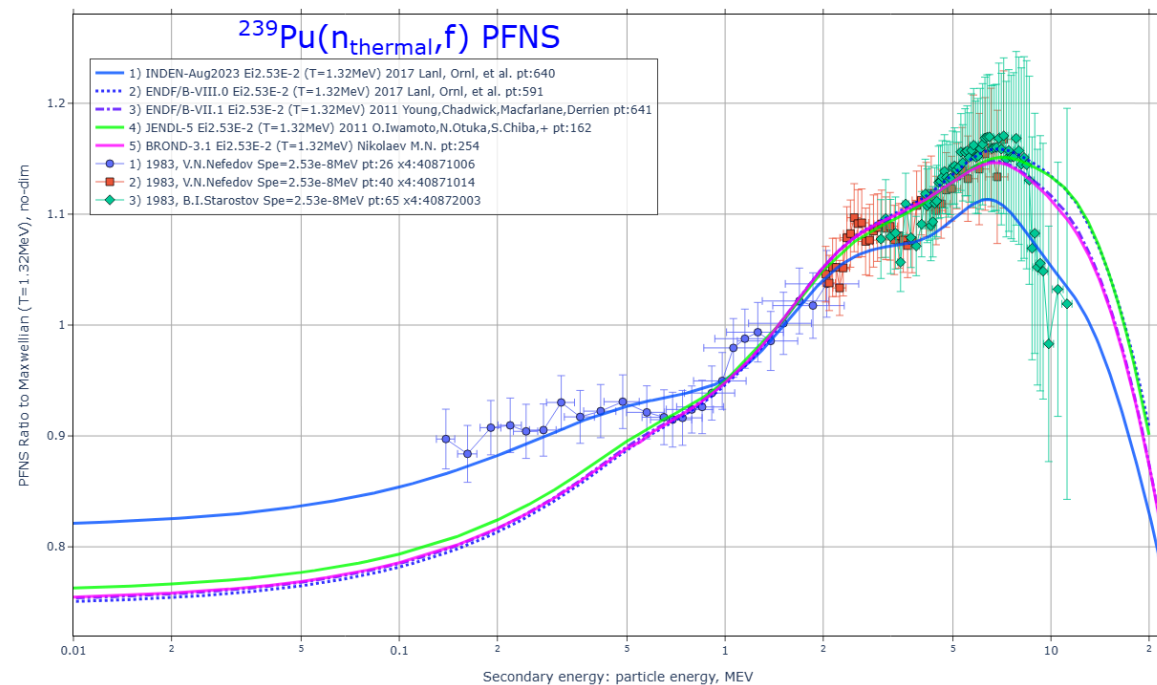
part2-B-pfns/ EXFOR/ENDF PFNS - energy spectrum of prompt fission neutrons

20) part2-B-pfns/out00/pfns233u_t.html.png $^{233}\text{U}(n_{\text{thermal}},f)$ PFNS-ratio to Mxw. (T=1.34MeV)

Reaction: 92-U-233(N,F),PR,NU/DE Quantity: MFQ:y=Data(Spe,E2) Datasets: 7 Points: 200
X4Pro, by V.Zerkin, Vienna, 2026, ver. 2026-09-13 //running: 2026-09-13 21:58:49

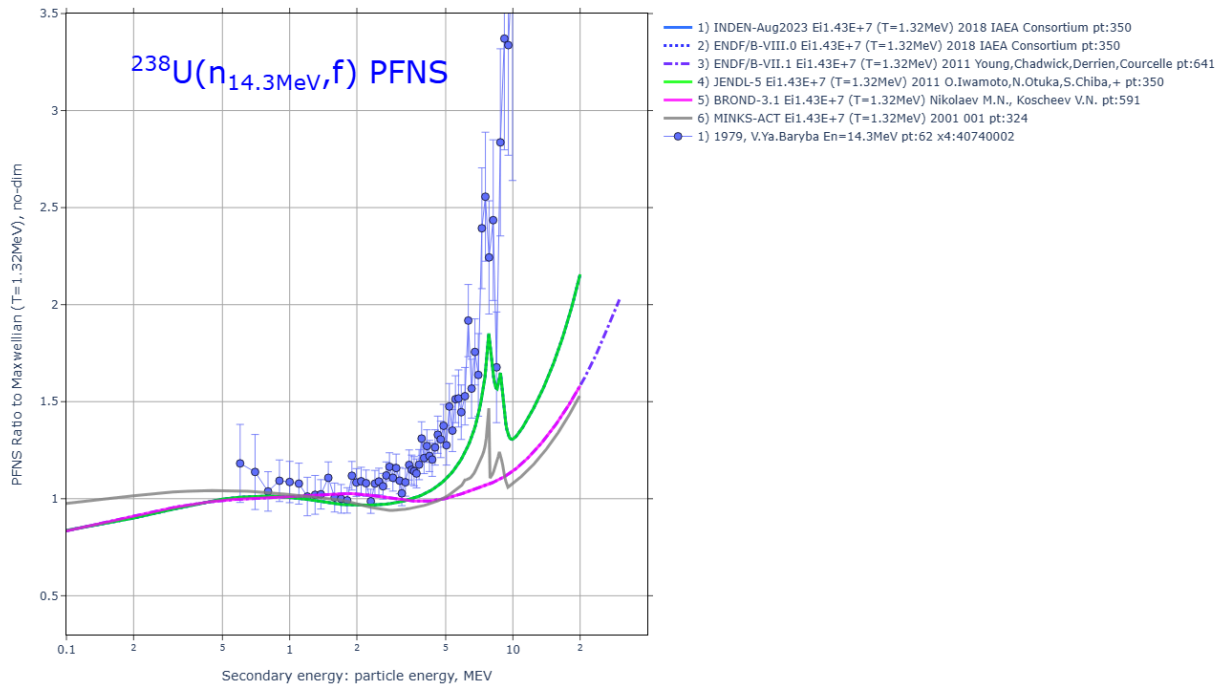
21) part2-B-pfns/out00/pfns239pu_t.html.png $^{239}\text{Pu}(n_{\text{thermal}},f)$ PFNS-ratio to Mxw.distr. (T=1.32MeV)

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver. 2026-09-02 //running: 2026-09-13 17:52:38



22) part2-B-pfns/out00/pfns238u14_3.html.png $^{238}\text{U}(n_{14.3\text{MeV}},f)$ PFNS-ratio to Mxw. (T=1.32MeV)

Reaction: 92-U-238(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 62
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running: 2026-09-13 17:52:18



23) part2-B-pfns/out00/pfns239pu1_45.html.png $^{239}\text{Pu}(n_{1.45\text{MeV}},f)$ PFNS-ratio to Mxw. (T=1.32MeV)

Reaction: 94-PU-239(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 3/4 Points: 136/165
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running: 2026-09-13 17:53:01

